

HCMO: A FAIR Ontology for Home-Cage Monitoring

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Abstract. Home-Cage Monitoring (HCM) systems observe laboratory animals continuously in their living environment, improving the contextual richness and reproducibility of preclinical data. Yet HCM systems are highly heterogeneous — differing in sensors, measured variables and output formats — and their data are inseparable from a rich experimental context. This heterogeneity hinders comparison, reuse, and FAIR data sharing. We present HCMO, the Home-Cage Monitoring Ontology, which builds on the COST TEATIME HCM Definition, and is the first machine-actionable ontology to represent HCM as integrated experimental and data-acquisition systems. HCMO is organised into five modules covering monitored enclosures, biological subjects, environmental context, observations and results, and technical acquisition. It is built around an explicit separation of sensor, observation, and result, preserving the chain from device to interpreted measurement. The published HCMO (v0.3.0) reuses BFO, IAO, [the 2017 SOSA/SSN Recommendation](#), schema.org, OWL-Time, [QUDT](#), and a reviewed SemTS time-series segment pattern where their semantics fit. HCMO is delivered as a FAIR,

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tool-consumable package: modular sources, reproducibly generated distributions (Turtle/OWL/JSON-LD), SHACL shapes, competency-question queries, a JSON-LD context, persistent identifiers, an open licence, and documentation. [We evaluate it with pitfall and FAIRness scanners, reasoning, SHACL validation, exact-answer competency questions, and an ISA/RO-Crate round-trip fixture.](#)

Keywords: ontology · home-cage monitoring · laboratory animals · FAIR data · SOSA/SSN · 3Rs

1 Introduction

Behavioural testing is central to preclinical research. [In rodents, it still relies heavily on short tests performed *outside* the animal’s home cage, in a novel environment \[19\]. Such protocols \[capture\]\(#\) only a narrow window into an animal’s activity, are frequently run during the light phase even though rodents are predominantly nocturnal, and require handling and transfer that introduce stress and behavioural bias. \[This limits the observation of circadian rhythms and slowly emerging changes, and contributes to variability across laboratories \\[19\\].\]\(#\)](#)

Home-Cage Monitoring (HCM) inverts this logic: instead of bringing the animal to the test, the observation is brought into the animal’s usual environment. Through sensors, hardware, and software, an HCM system continuously observes an animal living in a space that provides food, water, social contact, and safety, over days, weeks or months and with limited human intervention [19,18]. This yields more regular, better-contextualised data and supports the 3Rs (Replacement, Reduction, Refinement) — particularly Refinement — by reducing handling and handling-related stressors. [HCM does not, however, remove methodological bias so much as *shift* part of it toward sensors, analysis software, variable definitions, and metadata quality \[19\].](#)

This shift creates a data-interoperability problem. HCM systems vary widely — video, RFID, infrared and motion sensors, food- and water-intake tracking, or combinations thereof; commercial, lab-built, or hybrid [18,9]. Two studies may measure closely related behaviours while [relying on](#) very different sensors, parameters, and file formats. Crucially, HCM data are inseparable from their experimental context: a measurement is interpretable only if one also knows the animal (species, strain, sex, age, genotype), the enclosure and its environmental conditions, the device and its configuration, the time, and the protocol [10]. This heterogeneity hinders comparison, reuse, and integration, and stands in the way of FAIR (Findable, Accessible, Interoperable, Reusable) data [33]. In animal research, data quality is moreover an *ethical* concern, since the data are the durable result of animal experimentation [25].

Controlled vocabularies, taxonomies, thesauri, or metadata standards help, but they fall short for a domain whose value lies in the *relations* between an animal, an enclosure, a sensor system, a protocol, environmental conditions, observed behaviours, and provenance. An ontology formalises the concepts of the domain, specifies their relations, reuses existing identifiers, and makes the data

queryable, comparable, and integrable, while enabling consistency checking and inference [23].

We present **HCMO**, the Home-Cage Monitoring Ontology, which formalises HCM as an integrated system. Its contributions are:

1. **An ontology for the HCM domain** organised into five modules: a minimal enclosure-centred *core*, *bio* (subjects, groups, factors, and housing assignments), *env* (environmental profiles and measurements), *obs* (observations and results), and *tech* (hardware, sensors, actuators, software, and time series). The model is built around an explicit *sensor* $\neq$ *observation* $\neq$ *result* separation, and keeps the monitored subject (*bio*) and its environmental conditions (*env*) distinct from the observations made about them (*obs*).
2. **Selective, reviewed reuse of established standards** (BFO/IAO, SOSA/SSN, QUDT, SemTS, OWL-Time), with an explicit policy on how strongly each reuse is claimed, and an instance-level bridge to the ISA/OBI/STATO ecosystem.
3. **A FAIR, tool-consumable resource package**: a stable release manifest, modular Turtle sources, a reproducibly generated merged graph (TTL/OWL/JSON-LD), SHACL shapes, a JSON-LD context, competency-question SPARQL, HTML documentation, a persistent IRI, an open licence (CC BY 4.0), and a DOI.
4. **A competency-question-driven evaluation** of coverage and quality, including an executable ISA/RO-Crate round-trip use case.

The remainder of the paper positions HCMO against related work (Sect. 2), describes the development methodology (Sect. 3), presents the resource (Sect. 4) and its interoperability use case (Sect. 5), reports the evaluation and potential impact (Sect. 6), and discusses limitations and future work (Sect. 7).

2 Related work

COST Action TEATIME. HCMO originates in COST Action CA20135, *Improving biomedical research by automated behaviour monitoring in the animal home-cage* (TEATIME)⁸, and is developed with members of its network. The Action’s outputs frame the problem HCMO addresses: the edited volume *Home Cage Monitoring in Rodents: A Global Effort* [12], whose chapters survey available systems, sensing technologies, experimental applications, and do-it-yourself approaches [19,18,9] as well as HCM data sharing and metadata [10]; a minimal metadata set (MNMS) for repurposing nonclinical in vivo data [21]; and a FAIR-compliant repository architecture for HCM data and metadata that calls for standardised data structures, comprehensive metadata, and ontology-supported harmonisation [2]. The WellFAIR perspective adds that preserving the value and reusability of animal-derived data is part of responsible animal research [25].

⁸ <https://www.cost-teatime.org/>

Table 1. Positioning of HCMO against related resources. “Formal” means a deployable OWL/RDF artifact; “HCM chain” means the enclosure–subject–environment–device–observation–result path of continuous in-cage acquisition.

Resource	Scope	Formal	HCM chain
HCM Definition Olog [7]	What qualifies as HCM; reportable characteristics	–	conceptual
Bains et al. model [2]	Semantic components of HCM experiments and data	–	conceptual
OBI [3]	Investigations, protocols, instruments, data	✓	–
OLAM	Laboratory-animal medicine terminology	✓	–
MEDO	Mouse experimental design	✓	–
ARRIVE [31]	Reporting guideline for in-vivo studies	–	–
ISA [30] / RO-Crate [32]	Investigation metadata exchange / packaging	JSON-LD	–
HCMO (this work)	HCM experiments and data acquisition	✓	✓

Conceptual models of HCM. Efforts to define and structure HCM concepts predate HCMO. Within TEATIME Working Group 2, dedicated to the current status of HCM, the HCM Definition Olog emerged collectively and was refined through exchanges among its members [7]; it supports deciding whether a system qualifies as HCM and identifies the defining experimental characteristics that should be reported [7,19]. A related model, Fig. 1 of Bains et al. [2], developed from the Olog with a different objective: to enumerate the semantic components needed to describe and manage HCM experiments and their data. The Olog thus establishes the domain boundary, whereas the Bains et al. model identifies more detailed representation requirements. Both are graphical conceptual models rather than deployable Semantic Web ontologies; the Olog explicitly adopts a simplified Olog representation [7].

Adjacent biomedical and laboratory-animal resources. Several resources cover neighbouring concerns. The Ontology for Biomedical Investigations (OBI) models investigations, protocols, instruments, materials, and data [3]; the Ontology of Laboratory Animal Medicine (OLAM) provides laboratory-animal terminology; and the Mouse Experimental Design Ontology (MEDO) standardises experimental-design descriptions, overlapping HCMO’s housing/experimental-group notions. Reporting guidelines such as ARRIVE [31] prescribe *what* to report about in-vivo experiments but are not machine-readable ontologies. The ISA framework provides a widely used investigation–study–assay exchange model [30], while RO-Crate packages research data and contextual metadata as JSON-LD [32]; HCMO uses both as exchange targets (Sect. 5). Table 1 summarises the positioning.

Gap. Previous HCM data-sharing work establishes the need for standardised metadata, controlled terminology, and interoperable data structures [2,10], but does not provide a shared ontology capable of representing HCM experimental entities and their relationships in a machine-actionable form, and the adjacent ontologies do not model continuous in-cage acquisition. HCMO addresses this formalisation gap through an explicit semantic model of the HCM acquisition chain—linking animal subjects, housing and environmental context, devices and sensors, observations, derived measurements, behaviours, and results—with formally defined entities and relations intended for annotation, validation, in-

tegration, and computational reuse across heterogeneous HCM platforms; it is complementary to OBI and MEDO and can bridge to them where appropriate.

3 Methodology

Development process. HCMO follows an incremental, competency-question-driven process in the spirit of LOT [26] and SAMOD [24], with requirements expressed as competency questions [23]. The current version was conceptualised graphically in diagrams.net and exported to OWL/Turtle with Chowlk [6]; LinkML was explored as an alternative serialisation [22]. Documentation is produced with WIDOCO [14], and quality is assessed with OOPS! [27] and, for FAIRness, FOOPS! [15], complemented by reasoning and SHACL validation [13].

Use of AI assistants. HCMO is developed in its public GitHub repository, to which AI coding assistants (OpenAI Codex and Anthropic Claude Code) were connected. Working on branches, under repository-level agent instructions (AGENTS.md) that forbid silent changes of meaning, re-minted IRIs, and hand-edited generated files, they drafted changes to modules, shapes, queries, tooling, and documentation, ran the build and validation gates, and helped draft and restructure this manuscript. Their changes were reviewed by an author before merging and had to pass the same CI gate as human edits. Modelling decisions — scope, definitions, axioms, and alignments — were taken by the authors, often in co-author meetings; AI-generated reviews, such as the term-by-term assessment of proposed upper-level alignments, were treated as input to those decisions, not as decisions. The authors take full responsibility for the content.

Stakeholders. Researchers comparing data across studies, facility managers and welfare committees auditing husbandry, data engineers ingesting device exports, HCM manufacturers, and, later, AI consumers of the knowledge graph share one need: to interpret a measurement together with the context of its production.

Requirements. Analysis of the HCM domain [19,18,10] yields eight requirements (Table 2).

Competency questions. Selected retrieval aspects of the requirements are operationalised as executable SPARQL questions [23,26]; structural requiredness is tested separately with SHACL. One representative question is: *Which enclosure contains the reviewed animal at a stated reference time according to its time-bounded housing assignments?* The release index contains eleven questions covering assignments, dimensions, environment specifications, operational and calibration evidence, husbandry provisioning, sensor-captured properties, long observations, and ISA/STATO provenance paths. Complete expected bindings are versioned with the queries, including the intentionally empty missing-dimensions answer.

Selecting and reusing external resources. External vocabularies were chosen by role: SOSA/SSN for the sensor/observation/platform backbone [17], OWL-Time for temporal entities and intervals [8], QUDT for quantity values and units [28], PROV-O for provenance [20], BFO/IAO for upper-level grounding [1],

Table 2. Requirements derived from the domain analysis.

Requirement	Content
R1 Context-aware observations	An observation remains linkable, where applicable, to its feature of interest, observed property, time, procedure, sensor, and result; shapes enforce fields per profiled observation subtype rather than one completeness rule for every SOSA observation.
R2 Device $\neq$ observation $\neq$ result	A sensor, the measurement event in context, and the produced value or interpretation are distinct, preserving <i>device</i> $\rightarrow$ <i>observation</i> $\rightarrow$ <i>result</i> ; HCM outputs are often signals later transformed by software into inferred behaviours.
R3 Temporal housing	Assigning an animal to an enclosure is time-bounded, represented over an interval, and linked to experimental groups and study factors.
R4 Environment as context	Enclosure dimensions, bedding, enrichment, water/food, light cycles, and measurable conditions are representable, distinguishing a <i>property</i> from the <i>values</i> observed for it.
R5 Technical provenance	Hardware, sensors, software, and their organisation of data into time series/segments are captured without conflating technical provenance with biological interpretation.
R6 Rich metadata	Species, strain, sex, age, housing, enrichment, device configuration, sampling rate, software, and protocol are first-class, supporting FAIR reuse [33,10].
R7 Standards reuse	Suitable standards are reused rather than re-minted; each mapping strength is justified separately.
R8 Data-quality checking	The model supports consistency and completeness checks (e.g. a numeric value without a unit, an animal not assigned to an enclosure over an interval).

and SemTS for time-series segments and their dimensions [16]. A term is reused when (i) its published definition fits the HCM meaning, reviewed term by term rather than per vocabulary; (ii) it has a canonical, versioned artifact that HCMO can pin and checksum — hence the 2017 SOSA/SSN Recommendation rather than the evolving later edition; and (iii) its reasoning consequences are acceptable in the merged graph. Terms failing a review are removed rather than kept loosely (Sect. 4). We also distinguish two kinds of reuse: annotation properties used for ontology metadata (e.g. Dublin Core terms) and classes and properties that carry HCM semantics.

Claim strength. We distinguish implemented semantic reuse or alignment, validated interoperability evidence, and formal profile conformance. Implemented alignment requires canonical versioned terms, reviewed semantic fit, and a normative role in HCMO; its assertion strength is stated per term and never generalised to a whole vocabulary. **Interoperability evidence** (a validated, queried mixed-vocabulary example) establishes neither mappings nor complete round trips; formal conformance requires every applicable profile requirement to be tested for a declared version and scope.

4 The HCMO resource

Identity and size. HCMO uses the persistent ontology IRI <https://w3id.org/hcmo/ontology/hcm> and the stable term namespace <https://w3id.org/hcmo/ontology/hcm#>. Domain-specific terms use four sub-namespaces beneath the

Table 3. Summary of the HCMO 0.3.0 release. “Release” counts all HCMO-namespace terms in the generated distribution, including deprecated compatibility terms; “active” excludes them. Reused external terms are counted separately.

Resource measure	Release	Active
RDF triples in the canonical distribution	1,397	
HCMO classes	60	33
HCMO object properties	68	32
HCMO datatype properties	75	37
External classes/properties reused for HCM semantics	25	
SOSA 9, BFO 4, schema.org 4, SemTS 3, QUDT 3, IAO 1, PROV-O 1		
External vocabularies reused for ontology metadata		6
Authored modules	5 + compatibility	
Executable competency questions	11	

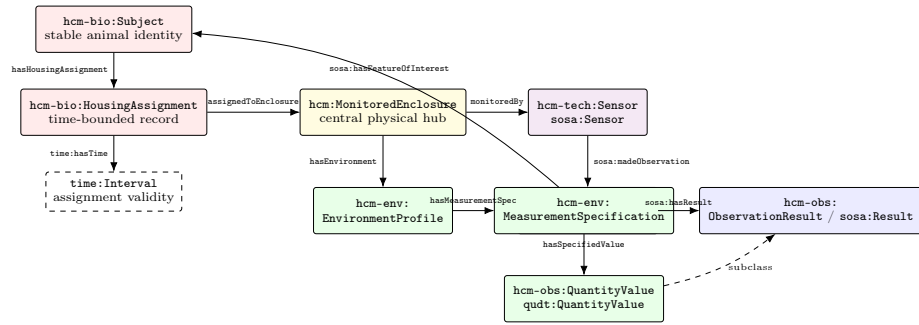


Fig. 1. The five-module HCMO model preserves animal identity, temporal housing, environmental specifications, sensing events, and results as distinct entities.

same base: `hcm/bio#`, `hcm/env#`, `hcm/obs#`, and `hcm/tech#`. Published IRIs are not renamed when terminology changes; obsolete IRIs are retained in a compatibility module with deprecation and replacement metadata where a defensible replacement exists. Table 3 summarises the release.

Modular organisation. HCMO 0.3.0 is authored as five active domain modules. The minimal core centres on `hcm:Enclosure`, `hcm:MonitoredEnclosure`, enclosure dimensions, enrichment, and stable housing relations. The *bio* module represents subjects, experimental groups, housing assignments, and study factors. The *env* module represents environmental profiles, light cycles, gas profiles, measurement specifications, and environmental properties. The *obs* module owns SOSA-style observations and their categorical, quantitative, behavioral, and tabular results. The *tech* module represents sensors, actuators, hardware, software, and time-series resources. Two further modules complete the manifest: an upper-level presentation module supplying the BFO/IAO anchors, and a migration-only module that preserves deprecated HCMO 0.0.1 IRIs but is not a source of terms for new data. Figure 1 gives a compact view of the principal entities and relations.

Key modelling decisions. This organisation preserves two boundaries that are important for HCM data. First, housing context is not reduced to the

animal: a housing assignment is an explicit record linking a subject or group to an enclosure. Second, a physical sensor, the observation it performs, and the result it produces are separate entities. For example, `hcm-tech:Sensor` is linked to a monitored enclosure, captures a `sosa:ObservableProperty`, and may make a `sosa:Observation`; the observation then identifies its feature of interest and result using SOSA relations. This supports provenance-sensitive queries without treating a device, event, and data value as the same thing. Environmental profiles, measurement specifications, and observations likewise use different relations: a profile composes specifications, each specification identifies the property and optional target quantity, and each observation uses the SOSA observed-property/result pattern. Housing membership is derived from time-bounded assignments at a stated time. Operational and calibration states are time-bounded records generated by explicit PROV activities rather than timeless booleans.

Reused standards. Physical subjects, enclosures, enrichments, sensors, actuators, hardware, and experimental groups are exposed under BFO material entity in the default end-user hierarchy. Records, specifications, profiles, software, and observation-result artifacts use the IAO information-content anchor. A separate optional developer profile restores BFO’s source-faithful intermediate hierarchy and the more precise object-aggregate classification of experimental groups. Sensors, actuators, observations, results, and observed properties reuse SOSA classes and relations, and `hcm-tech:captures` is explicitly a subproperty of `sosa:observes`. HCMO normatively follows the 2017 SOSA/SSN Recommendation [17], pinned to an immutable copy of the 2017 ontology artifact; terms introduced by the developing later edition are not mixed into this release. Numeric dimensions, sampling rates, environmental targets, and observation results use the pinned QUDT 3.4.0 `QuantityValue`, `numericValue`, and `hasUnit` pattern [28], with HCMO role properties stating what each quantity describes. HCMO also selectively reuses canonical SemTS 1.2.0 terms [16]: a location result table is a `semts:TimeSeriesSegment` whose dimensions may be linked with `semts:segmentDimension` to `semts:DataDimension`; the earlier observation dimension restriction and `semts:generated` relation were removed after semantic review. Schema.org supplies contributor and place exchange types, and the example ABox uses OWL-Time intervals. These are selective references to pinned artifacts, not claims that HCMO imports or reproduces the full external ontologies.

Engineering and distribution. HCMO is released as a **tool-consumable package**. A machine-readable manifest (`hcmo.yaml`) declares the ontology’s identity and version IRIs, the ordered source modules, the generated distributions, the SHACL shapes, the competency-question index, and the example datasets; its *shape* is treated as an API and kept stable, so consumers discover paths from the manifest rather than hard-coding them. A build step merges the seven Turtle sources into a canonical, sorted serialisation that is byte-identical on re-run, published as Turtle, RDF/XML, and JSON-LD together with a flat term inventory (`profile.json`) for synchronisation layers and interfaces, and

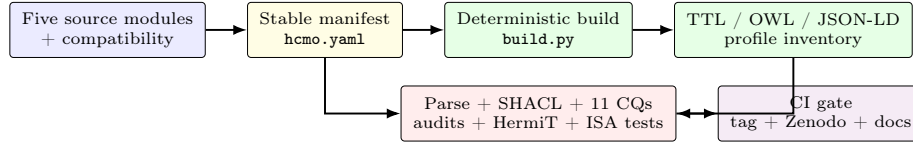


Fig. 2. HCMO’s manifest-driven, reproducible build and validation pipeline.

a JSON-LD context (`ontology/context.jsonld`) for application developers. The presentation module adds a second `owl:Ontology` node; metadata scanners should select the root header (version IRI, licence, DOI, creator ORCID) by its IRI. A validation step parses every file, runs the SHACL shapes against positive and intentionally invalid examples, and executes every competency question against complete expected answers; it gates every pull request in CI, and a tag-triggered workflow publishes the release (Fig. 2).

Availability and sustainability. The ontology and its terms dereference from the w3id IRI via HTTP 303 content negotiation to WIDOCO documentation [14] (<https://dhuzard.github.io/HCMO/index-en.html>). Sources and distributions are openly available from the public GitHub repository and the Zenodo deposit (release v0.3.0, DOI 10.5281/zenodo.22208202). No hosted SPARQL service is part of HCMO 0.3.0. The distributions can be loaded into any standards-compliant triple store. The resource is licensed CC BY 4.0, carries provenance on the root ontology header (creators with ORCID, version IRI), and provides a canonical citation (`CITATION.cff`); a FAIR self-assessment is available in the repository. HCMO is maintained through public issues and pull requests; versioning follows SemVer mirrored by `owl:versionIRI`, and terms are deprecated rather than deleted or re-minted to protect external references. HCMO is registered in BioPortal (<https://bioportal.bioontology.org/ontologies/HCMO>).

DH: update the BioPortal entry with the final release before submission.

5 Interoperability use case

HCM study metadata travel through exchange formats such as ISA [30], and their statistics are described with OBI and STATO. HCMO does not import or align OBI and STATO as HCMO classes; the release shows at the instance level how HCMO data connect to that ecosystem, in two steps (Fig. 3).

A recording-to-result evidence slice. A pinned example, `examples/isa-hcmo-bridge.ttl`, uses PROV-O, specific OBI and STATO types, and ISA/Bioschemas exchange terms together with HCMO in one acyclic workflow from a raw recording to a statistical result [30,32]. This is validated interoperability evidence, not an HCMO class mapping or a claim of formal ISA RO-Crate conformance.

A 2 × 2 factorial study with repeated measures. The second step is a complete, generated study. Eight individually housed animals are allocated to a 2 × 2 factorial design — treatment (vehicle or active) crossed with cage enrichment (standard or enriched) — giving four groups of two. Each animal’s dark-phase activity is observed on seven consecutive days, so day is a repeated, within-animal

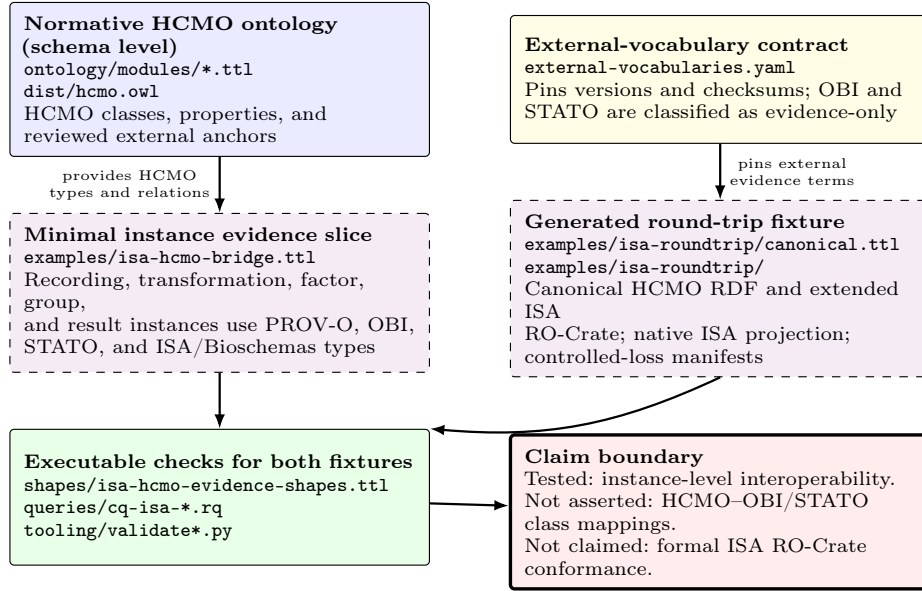


Fig. 3. Separation between HCMO’s normative ontology and its instance-level interoperability evidence. OBI, STATO, and ISA/Bioschemas occur in the linked example fixtures and checks; their use there is not an HCMO class mapping or a formal ISA conformance claim. Repository-relative paths identify the executable artifacts behind each claim.

measure rather than a third crossed factor (56 observations in total), and one animal is re-housed during the study. A 2×2 design is the smallest one that exercises two crossed factors, their groups, and an interaction; the repeated measures and the re-housing test that observations stay attached to the right animal and enclosure over time. The fixture asks whether this structure — animal identity, housing history, factors, groups, repeated observations, a derived tissue sample, and the statistical results — survives conversion between HCMO and ISA, and what is lost when it does not.

The fixture contains non-overlapping re-housing, an actual derived tissue bioschemas:Sample, and separate fitted-model, estimate, confidence-interval, p-value, schema:MediaObject, and file-fragment entities. Dark-phase outcomes use phenomenon intervals, and the re-housed animal’s observations resolve to its new enclosure through the authoritative assignment. The declared mixed model is executed against the generated CSV with pinned numerical dependencies; the model serialization and the active-versus-vehicle contrast at standard enrichment use distinct exact row fragments. The input matrix (dark-phase-activity.csv), the STATO-annotated result matrix (model-results.csv), the STATO-typed entities and exact fragment links (canonical.ttl), and an artifact guide (README.md) are in the repository.

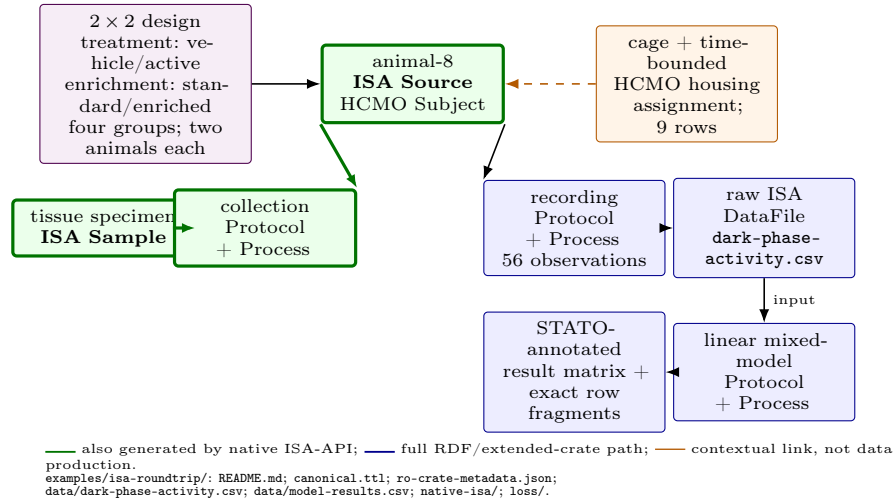


Fig. 4. Executable 2×2 factorial fixture with repeated measures (seven daily observations per animal). Each entity and output has a distinct role. The green animal Source $\rightarrow$ collection Process $\rightarrow$ genuine tissue Sample path is reproducibly generated as ISA-JSON and ISA-Tab. The remaining factor, housing, whole-animal recording, repeated-observation, and semantic STATO/file-fragment structures are validated in HCMO RDF and the extended ISA RO-Crate, but remain declared losses in the native ISA projection. The result CSV exposes the reviewed STATO IRIs and labels while the RDF retains the typed entities and exact row-fragment links.

Three projections, two levels of fidelity. Canonical HCMO RDF and an extended ISA RO-Crate JSON-LD serialisation are graph-isomorphic at 1,539 triples. `roc-validator` 0.11.3 passes all RO-Crate 1.2 required rules and all ISA-specific required rules; its full inherited ISA run has one isolated upstream conflict, since the embedded ISA profile requires RO-Crate 1.1 although its minimal ISA fixture uses the 1.2 context. Formal ISA conformance therefore remains deferred: the draft profile has no permanent URI and its validator disagrees about the base RO-Crate edition. Native ISA-JSON and ISA-Tab are controlled-loss projections. Pinned `isatools` 0.14.3 reproducibly generates checked-in ISA-JSON and ISA-Tab files and validates their JSON-Tab-JSON round trip for the genuine Source-to-Sample collection: a distinct animal Source, a real tissue Sample, their collection process, and their HCMO identifiers, which survive as ISA comments. Housing, direct whole-animal recording, source-bound factor assignments, explicit groups, repeated observations, and semantic STATO/file-fragment links remain declared losses, checked against loss manifests, rather than being represented with fabricated Sample proxies (Fig. 4). The generated files and modelling decisions are documented under `examples/isa-roundtrip/native-isa/`.

Table 4. [Ontology-quality scans of the 0.3.0 RDF/XML artifact. Raw outputs and term-level triage are archived in the repository.](#)

Tool	Run	Outcome
FOOPS! [15]	15 Aug; 31 Aug 2026 (release)	Score 1.0; all metadata, licensing, provenance, resolvability, vocabulary-reuse, label, and definition tests pass; byte-identical response on the release.
OOPS! [27]	15 Aug; 31 Aug 2026 (release)	P04, P08, P10, P13, P22, P34, P35; same codes and counts on the release (trriage below).
AskWol [11]	after deployment	Namespaces resolved; domain/range, datatype, licence, IRI-scheme, and reasoner checks pass; no contradiction or unsatisfiable class. Informational only (beta).

6 Evaluation and potential impact

HCMO is evaluated at four complementary levels: common ontology pitfalls and FAIR metadata, OWL consistency, instance-data validation, and executable competency questions. [All reports, raw outputs, checksums, and environments are archived in the repository.](#)

Pitfalls and FAIRness. Table 4 summarises the scans. The important OOPS! P10 finding is the scanner’s ontology-wide missing-disjointness heuristic and names no affected pair. P34 and P35 concern reused SOSA, SemTS, QUDT, Schema.org, and PROV terms whose authoritative declarations are covered by HCMO’s pinned external-vocabulary contract rather than copied into the merged graph; P04, P08, and P22 likewise concern reused BFO or external terms. P13 marks 32 properties as lacking inverse declarations; candidate inverses are not asserted without definition-level evidence, because doing so would create false entailments merely to satisfy a scanner. AskWol’s mixed-strategy warning reflects HCMO’s stable core-plus-module namespace architecture; the IRIs are not re-minted to satisfy it. Triage reports: OOPS!/FOOPS! and AskWol.

Logical consistency. The canonical RDF/XML artifact was rebuilt and checksummed again on 24 August 2026 after a metadata-only DOI correction. HermiT loaded 65 classes and reported zero inconsistent classes. The pre-check found no active UNKNOWN: IRI and no property declared as both object and datatype property.

SHACL validation. The validator supplies the canonical ontology as a separate ontology graph and enables RDFS entailment when validating each example, so that domain, range, and subclass consequences select the same `sh:targetClass` constraints as explicit types. Three positive examples conform, while two edge-case examples produce their reviewed violations. One negative fixture deliberately omits explicit `rdf:type`: it conforms when validated without the ontology and fails when ontology-aware inference selects the intended targets. This keeps profile requiredness in SHACL rather than in OWL existential semantics. A separate ISA/STATO evidence shape validates the workflow boundary of Sect. 5; an injected process/data cycle is required to fail. [For the round-trip fixture, dedicated shapes and injected probes reject inconsistent housing assignments,](#)

replacement animals, observation/housing and group/factor mismatches, and fabricated ISA assignment results.

Competency questions. Eleven SPARQL queries run over the ontology plus all positive examples, and the validator checks the complete expected answer rows, including bindings, unbound values, and multiplicity. For the representative temporal-housing question of Sect. 3, the query returns the reviewed subject, enclosure, and housing-assignment IRI at the stated reference time, exactly matching its single expected answer. All eleven queries match their complete versioned answer sets. The zero-row dimensions query is intentional: it verifies the absence of missing records rather than being accepted as an unexamined empty result. Five further exact-answer questions run against the isolated round-trip fixture of Sect. 5: nine housing-history rows, eight authoritative subject/group/factor rows, eight per-animal repeated-observation counts, four statistical-result/file-fragment rows including exact values and meaning links, and one Source-to-derived-Sample row.

Potential impact. HCM exports are often weakly linked to their subject, enclosure, environment, device, and protocol [2,10,25]. By making the provenance and experimental context of each measurement explicit, HCMO can support dataset discovery, cross-system comparison, integration, and reuse, provided that datasets are mapped to the ontology and exposed with appropriate identifiers and metadata. In animal research, where data are the durable output of an experiment, preserving their context also has ethical relevance: it can reduce avoidable duplication, enable secondary analyses, and increase the scientific value obtained from each animal, supporting Reduction and Refinement [25]. For AI-supported analysis, it supplies admissible types and relations that can be enforced through validation, ontology-aware retrieval, and type-constrained query generation, though it does not by itself prevent erroneous inference.

Adoption path. Development is grounded in a network of preclinical and HCM experts originating in COST Action TEATIME, which provides domain feedback and a route to real datasets for validation and uptake. Because the release is recent, longitudinal user and deployment statistics are not yet available; adoption is therefore presented as a pathway rather than a demonstrated outcome.

7 Discussion, limitations, and future work

The evaluation shows that HCMO is consistent, FAIR-scored, and executable, with shapes, questions, and fixture checked on every change. It does not establish formal alignments, formal ISA conformance, or community adoption.

Limitations. HCMO is still an early and evolving resource. Some conceptual boundaries are being stabilised, most notably between raw data, processed measurement, interpreted observation, and biological result, where the same information can be viewed at several levels. The diversity of HCM systems makes a single model that is both broad enough to be reusable and precise enough to describe individual experiments an ongoing balance. Statistics expose this sharply: some systems produce measurements only per cage or group, others per individual

animal, so HCMO cannot yet define the experimental unit unambiguously, nor link both cases to ISA in a mutually compatible way. Alignment to external vocabularies is deliberately partial in this version, and the ISA/OBI/STATO connection is instance-level evidence, not an HCMO class mapping. The factorial and mixed-model resources are interoperability fixtures, not a general HCMO statistics hierarchy.

Future work. Planned directions include expert review of definitions and axioms identified by the class and property audits; systematic alignment decisions (reuse as-is vs specialise vs keep HCMO-specific) and bridge modules to adjacent ontologies (e.g. OBI, PROV-O, STATO, MEDO) and to taxa/anatomy/device vocabularies; extending the current selective QUDT pattern to additional quantity roles where justified; and populating the ontology with real datasets via the TEATIME network and publishing the corresponding SPARQL queries. HCMO is also designed to underpin tooling that is not a contribution of this paper: *knowledge-graph question answering* with ontology-checked queries [29,4,5]; *assisted authoring* through forms generated from the ontology and its shapes; and *vendor-data mapping* of proprietary exports into HCMO with preserved provenance.

Conclusion. HCMO is, to our knowledge, the first ontology to model home-cage monitoring as an integrated system spanning the animal, its housing, its environment, and the technical acquisition chain, delivered as a pinned, validated, and citable resource grounded in the COST TEATIME community. Through community review and the steps above, it aims to become a reference that improves the annotation, comparison, and reuse of HCM data across laboratories.

Resource Availability Statement. HCMO 0.3.0 — the modular Turtle sources, the merged Turtle, RDF/XML and JSON-LD distributions, the JSON-LD context, and the release manifest — is available under CC BY 4.0 from its persistent IRI <https://w3id.org/hcmo/ontology/hcm>, from the GitHub repository at tag v0.3.0 (<https://github.com/dhuzard/HCMO/tree/v0.3.0>), and from Zenodo (<https://doi.org/10.5281/zenodo.22208202>); the canonical citation is given in CITATION.cff. The SHACL shapes, example datasets including the 2×2 round-trip fixture, executable SPARQL competency questions with their expected answers, build and validation tooling, evaluation reports, and the FAIR self-assessment are in the same repository and archive. HTML documentation is at <https://dhuzard.github.io/HCMO/index-en.html>. No resource is subject to access restrictions.

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DH: the paper will describe the next release (v0.4). Before submission, update all version numbers, counts, DOIs, dates, and links to v0.4; the fixture README and native-isa/ files (post-0.3.0) are currently unreachable on GitHub main.

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Provenance. HCMO builds on the community-developed HCM Definition Olog (Sect. 2). From March 2025, Damien Huzard developed an earlier machine-actionable ontology, HCM-D, published on BioPortal and no longer maintained, within Metadatapp’s semantic-development work. The COST mobility-grant phase supported preliminary device-metadata mapping, project scoping, and an early GitHub version in September 2025. From October 2025 to October 2026, the present author team comprehensively redesigned and formally evaluated HCMO, producing the resource reported here.

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Competing interests. Damien Huzard initiated Metadatapp as an entrepreneurial project and currently maintains it as open-source software. He also provides consultancy services in preclinical research and research metadata through Neuronautix (<https://www.neuronautix.com>). The other authors declare no competing interests.

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